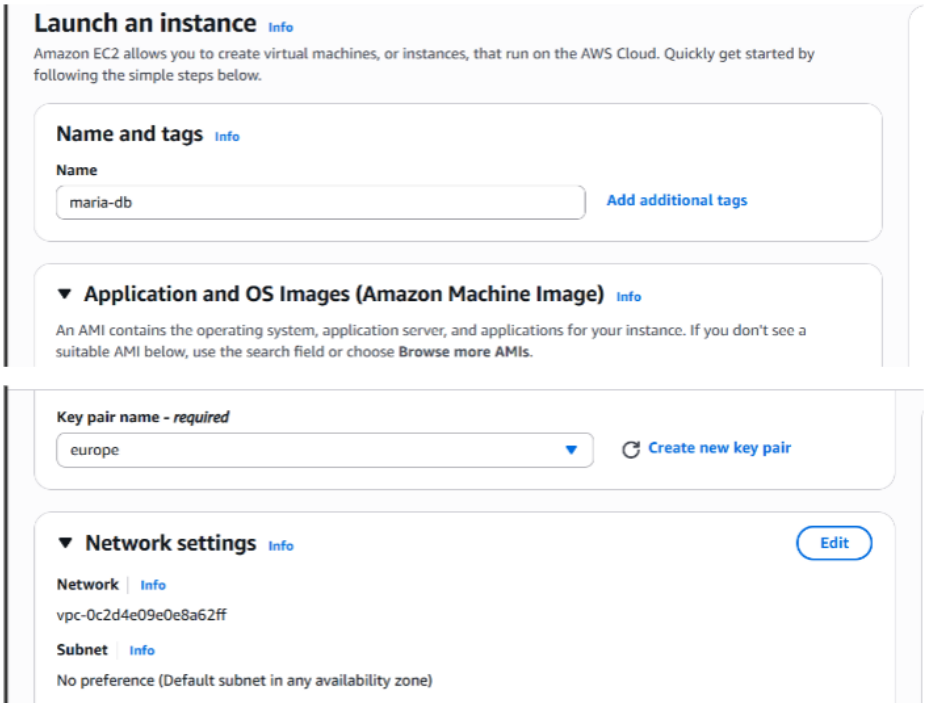
Relational Database Service:

1. Create MariaDB DB on EC2.

   First lets create an EC2 instance and install maria db



Updating the packages

Installing the maria db

```
[ec2-user@ip-172-31-42-2 ~]$ sudo dnf install mariadb105-server -y
Last metadata expiration check: 0:04:59 ago on Wed Apr 29 10:16:52 2026.
Dependencies resolved.
================================================================================================================
 Package                                    Architecture      Version                                  Repository
================================================================================================================
Installing:
 mariadb105-server                          x86_64            3:10.5.29-1.amzn2023.0.1                 amazonlinux
Installing dependencies:
 mariadb-connector-c                        x86_64            3.3.10-1.amzn2023.0.1                    amazonlinux
 mariadb-connector-c-config                 noarch            3.3.10-1.amzn2023.0.1                    amazonlinux
 mariadb105                                 x86_64            3:10.5.29-1.amzn2023.0.1                 amazonlinux
 mariadb105-common                          x86_64            3:10.5.29-1.amzn2023.0.1                 amazonlinux
 mariadb105-errmsg                          x86_64            3:10.5.29-1.amzn2023.0.1                 amazonlinux
 mysql-selinux                              noarch            1.0.4-2.amzn2023.0.3                     amazonlinux
 perl-DBD-MariaDB                           x86_64            1.22-1.amzn2023.0.4                      amazonlinux
 perl-DBI                                   x86_64            1.643-7.amzn2023.0.3                     amazonlinux
 perl-File-Copy                             noarch            2.34-477.amzn2023.0.7                    amazonlinux
 perl-Math-BigInt                           noarch            1:1.9998.39-2.amzn2023.0.2               amazonlinux
 perl-Math-BigRat                           noarch            0.2624-500.amzn2023.0.2                  amazonlinux
 perl-Math-Complex                          noarch            1.59-477.amzn2023.0.7                    amazonlinux
 perl-Sys-Hostname                          x86_64            1.23-477.amzn2023.0.7                    amazonlinux
Installing weak dependencies:
 mariadb105-backup                          x86_64            3:10.5.29-1.amzn2023.0.1                 amazonlinux
 mariadb105-cracklib-password-check         x86_64            3:10.5.29-1.amzn2023.0.1                 amazonlinux
```

```
[ec2-user@ip-172-31-42-2 ~]$ sudo systemctl start mariadb
[ec2-user@ip-172-31-42-2 ~]$ sudo systemctl enable mariadb
Created symlink /etc/systemd/system/mysql.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/mysqld.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/multi-user.target.wants/mariadb.service → /usr/lib/systemd/system/mariadb.service.
[ec2-user@ip-172-31-42-2 ~]$ sudo systemctl status mariadb
● mariadb.service - MariaDB 10.5 database server
     Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: disabled)
     Active: active (running) since Wed 2026-04-29 10:22:41 UTC; 28s ago
       Docs: man:mariadbd(8)
             https://mariadb.com/kb/en/library/systemd/
   Main PID: 26852 (mariadbd)
     Status: "Taking your SQL requests now..."
      Tasks: 13 (limit: 1014)
     Memory: 66.4M
        CPU: 464ms
     CGroup: /system.slice/mariadb.service
             └─26852 /usr/libexec/mariadbd --basedir=/usr

Apr 29 10:22:41 ip-172-31-42-2.eu-north-1.compute.internal mariadb-prepare-db-dir[26808]: The second is mysql@localhost, it has no p
Apr 29 10:22:41 ip-172-31-42-2.eu-north-1.compute.internal mariadb-prepare-db-dir[26808]: you need to be the system 'mysql' user to
Apr 29 10:22:41 ip-172-31-42-2.eu-north-1.compute.internal mariadb-prepare-db-dir[26808]: After connecting you can set the password,
Apr 29 10:22:41 ip-172-31-42-2.eu-north-1.compute.internal mariadb-prepare-db-dir[26808]: able to connect as any of these users with
Apr 29 10:22:41 ip-172-31-42-2.eu-north-1.compute.internal mariadb-prepare-db-dir[26808]: See the MariaDB Knowledgebase at https://m
```

2. Insert some Dummy data

Lets first connect to our data base

Setting environment variables and logging into db

```
[root@ip-172-31-42-2 ec2-user]# DBName=ec2db
[root@ip-172-31-42-2 ec2-user]# DBPassword=admin123
[root@ip-172-31-42-2 ec2-user]# DBRootPassword=admin123
[root@ip-172-31-42-2 ec2-user]# DBUser=ec2dbuser
[root@ip-172-31-42-2 ec2-user]# mysql -u root --password="${DBRootPassword}"
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 12
Server version: 10.5.29-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]>
```

Adding values in our database

```
MariaDB [(none)]> USE ec2db;
Database changed
MariaDB [ec2db]> CREATE TABLE table1 (id INT, name VARCHAR(45));
Query OK, 0 rows affected (0.008 sec)

MariaDB [ec2db]> INSERT INTO table1 VALUES(1, 'Virat'), (2, 'ABD'), (3, 'Rajat'), (4, 'Hazlewood');
Query OK, 4 rows affected (0.002 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

3. Take backup of dummy data on EC2

```
[root@ip-172-31-42-2 ec2-user]# mysqldump -u root -p ec2db > ec2db_backup.sql
Enter password:
[root@ip-172-31-42-2 ec2-user]# ls
ec2db_backup.sql
[root@ip-172-31-42-2 ec2-user]#
```

Using mysqldump utility to take backup of maria db.

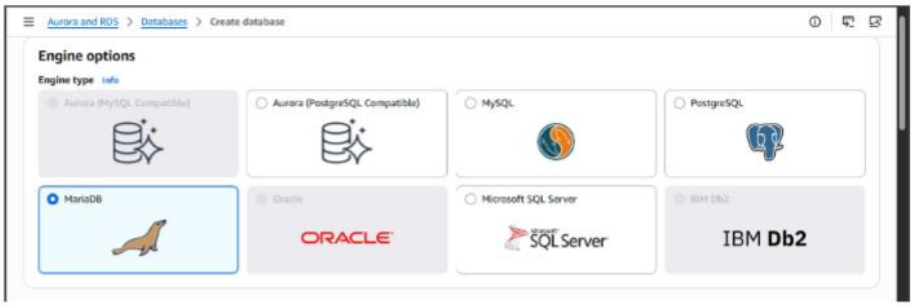
4. Launch Maria DB RDS instance

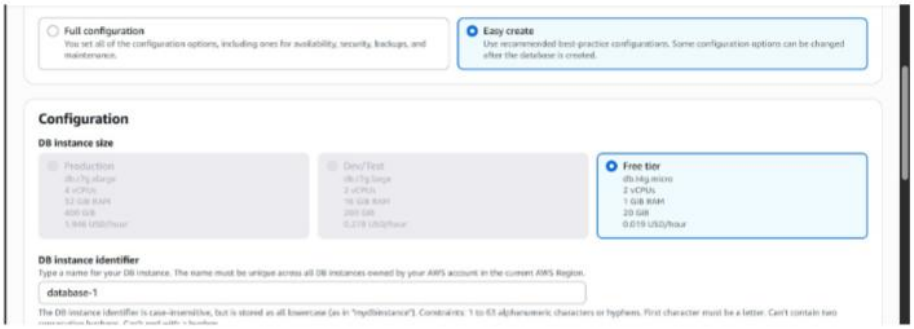
Navigate to RDS service

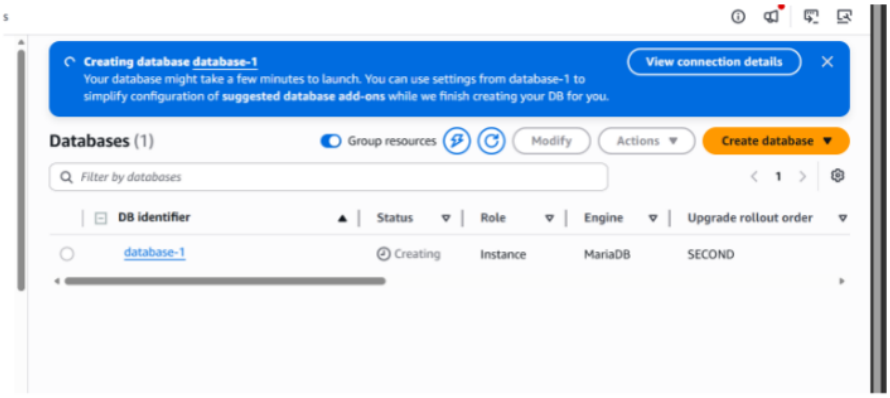
Create with full configuration

Create a database with custom configurations and enable features

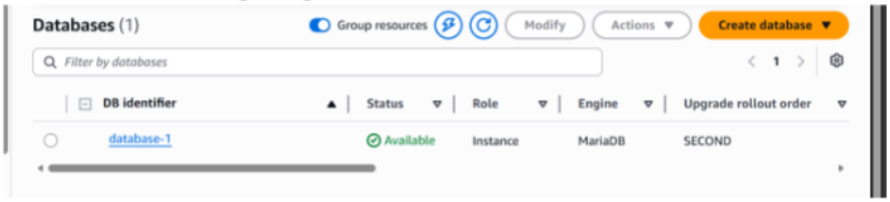
Click on create







Our database is now getting created



Database is available now

```
[root@ip-172-31-42-2 ec2-user]# mysql -h database-1.cfmgmyusmalj.eu-north-1.rds.amazonaws.com -u admin -p --ssl
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 60
Server version: 11.8.6-MariaDB-log managed by https://aws.amazon.com/rds/

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]>
```

5. Migrate database from EC2 to RDS

```
[root@ip-172-31-42-2 ec2-user]# ls
ec2db_backup.sql
[root@ip-172-31-42-2 ec2-user]#
```

We have a backup of previous data base already

```
[root@ip-172-31-42-2 ec2-user]# mysql -h database-1.cfmgmyusmalj.eu-north-1.rds.amazonaws.com -u admin -p --ssl
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 80
Server version: 11.8.6-MariaDB-log managed by https://aws.amazon.com/rds/

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]>
```

We are in our RDS database now

Creating a database

```
Type 'help;' or '\h' for help. Type '\c' to clea

MariaDB [(none)]> CREATE DATABASE ec2db;
Query OK, 1 row affected (0.015 sec)
```

```
MariaDB [(none)]> USE ec2db;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MariaDB [ec2db]> SHOW TABLES;
+-----------------+
| Tables_in_ec2db |
+-----------------+
| table1          |
+-----------------+
1 row in set (0.000 sec)

MariaDB [ec2db]> SELECT * FROM table1;
+------+-----------+
| id   | name      |
+------+-----------+
|    1 | Virat     |
|    2 | ABD       |
|    3 | Rajat     |
|    4 | Hazlewood |
+------+-----------+
4 rows in set (0.001 sec)
```

We can see that the database has been migrated from EC2 to RDS

6. Install MYSQL DB on EC2
   Downloading MYSQL

```
[root@ip-172-31-42-2 ec2-user]# wget https://dev.mysql.com/get/mysql80-community-release-el9-1.noarch.rpm
--2026-04-29 11:51:25--  https://dev.mysql.com/get/mysql80-community-release-el9-1.noarch.rpm
Resolving dev.mysql.com (dev.mysql.com)... 92.123.205.252, 2a02:26f0:41:694::2e31, 2a02:26f0:41:691::2e31
Connecting to dev.mysql.com (dev.mysql.com)|92.123.205.252|:443... connected.
HTTP request sent, awaiting response... 302 Moved Temporarily
Location: https://repo.mysql.com//mysql80-community-release-el9-1.noarch.rpm [following]
--2026-04-29 11:51:25--  https://repo.mysql.com//mysql80-community-release-el9-1.noarch.rpm
Resolving repo.mysql.com (repo.mysql.com)... 23.210.253.96, 2a02:26f0:e2:780::1d68, 2a02:26f0:e2:782::1d68
Connecting to repo.mysql.com (repo.mysql.com)|23.210.253.96|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 10534 (10K) [application/x-redhat-package-manager]
Saving to: 'mysql80-community-release-el9-1.noarch.rpm'

mysql80-community-release-el9-1.n 100%[===================================================>]  10.29K  --.-KB/s    in 0s
```

```
[root@ip-172-31-42-2 ec2-user]# sudo dnf install mysql-community-server -y --nogpgcheck
Last metadata expiration check: 0:05:45 ago on Wed Apr 29 11:55:51 2026.
Dependencies resolved.
================================================================================================================
 Package                                    Architecture         Version                     Repository
================================================================================================================
Installing:
 mysql-community-server                     x86_64               8.0.46-1.el9                mysql80-community
Installing dependencies:
 mysql-community-client                     x86_64               8.0.46-1.el9                mysql80-community
 mysql-community-client-plugins             x86_64               8.0.46-1.el9                mysql80-community
 mysql-community-common                     x86_64               8.0.46-1.el9                mysql80-community
 mysql-community-icu-data-files             x86_64               8.0.46-1.el9                mysql80-community
 mysql-community-libs                       x86_64               8.0.46-1.el9                mysql80-community

Transaction Summary
================================================================================================================
Install  6 Packages

Total size: 59 M
Installed size: 337 M
Downloading Packages:
[SKIPPED] mysql-community-client-8.0.46-1.el9.x86_64.rpm: Already downloaded
[SKIPPED] mysql-community-client-plugins-8.0.46-1.el9.x86_64.rpm: Already downloaded
[SKIPPED] mysql-community-common-8.0.46-1.el9.x86_64.rpm: Already downloaded
[SKIPPED] mysql-community-icu-data-files-8.0.46-1.el9.x86_64.rpm: Already downloaded
[SKIPPED] mysql-community-libs-8.0.46-1.el9.x86_64.rpm: Already downloaded
[SKIPPED] mysql-community-server-8.0.46-1.el9.x86_64.rpm: Already downloaded
Running transaction check
Transaction check succeeded.
Running transaction test
```

```
sudo systemctl start mysqld
sudo systemctl status mysqld
● mysqld.service - MySQL Server
     Loaded: loaded (/usr/lib/systemd/system/mysqld.service; enabled; preset: disabled)
     Active: active (running) since Wed 2026-04-29 12:03:19 UTC; 63ms ago
       Docs: man:mysqld(8)
             http://dev.mysql.com/doc/refman/en/using-systemd.html
    Process: 31331 ExecStartPre=/usr/bin/mysqld_pre_systemd (code=exited, status=0/SUCCESS)
   Main PID: 31358 (mysqld)
     Status: "Server is operational"
      Tasks: 38 (limit: 1014)
     Memory: 377.9M
        CPU: 1.606s
     CGroup: /system.slice/mysqld.service
             └─31358 /usr/sbin/mysqld

Apr 29 12:03:17 ip-172-31-42-2.eu-north-1.compute.internal systemd[1]: Starting mysqld.service - MySQL Server...
Apr 29 12:03:19 ip-172-31-42-2.eu-north-1.compute.internal systemd[1]: Started mysqld.service - MySQL Server.
[root@ip-172-31-42-2 ec2-user]#
```

```
[root@ip-172-31-42-2 ec2-user]# sudo mysql_secure_installation

Securing the MySQL server deployment.
```

```
New password:

Re-enter new password:

Estimated strength of the password: 50
Do you wish to continue with the password provided?(Press y|Y for Yes, any other key for No) : Y
By default, a MySQL installation has an anonymous user,
allowing anyone to log into MySQL without having to have
a user account created for them. This is intended only for
testing, and to make the installation go a bit smoother.
You should remove them before moving into a production
environment.

Remove anonymous users? (Press y|Y for Yes, any other key for No) : y
Success.


Normally, root should only be allowed to connect from
'localhost'. This ensures that someone cannot guess at
the root password from the network.

Disallow root login remotely? (Press y|Y for Yes, any other key for No) : y
Success.

By default, MySQL comes with a database named 'test' that
anyone can access. This is also intended only for testing,
and should be removed before moving into a production
environment.
```

```
[root@ip-172-31-42-2 ec2-user]# mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 13
Server version: 8.0.46 MySQL Community Server - GPL

Copyright (c) 2000, 2026, Oracle and/or its affiliates.
```


```
[root@ip-172-31-42-2 ec2-user]# mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 13
Server version: 8.0.46 MySQL Community Server - GPL
```

```
New password:

Re-enter new password:

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Do you wish to continue with the password provided?(Press y|Y for Yes, any other key for No) : Y
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Copyright (c) 2000, 2026, Oracle and/or its affiliates.
```


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Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 13
Server version: 8.0.46 MySQL Community Server - GPL

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```


```
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affiliates. Other names may be trademarks of their respective
```

```
mysql> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
4 rows in set (0.00 sec)
```

7. Launch MySQL RDS image

   Go to RDS service
   And

Create with full configuration

Create a database with custom configurations and enable features you want across a variety of engines including Aurora (compatible with MySQL & PostgreSQL), MySQL, PostgreSQL, MariaDB, Oracle, Microsoft SQL Server, and IBM Db2. Or use an existing database using restore from S3.

Create Restore from S3

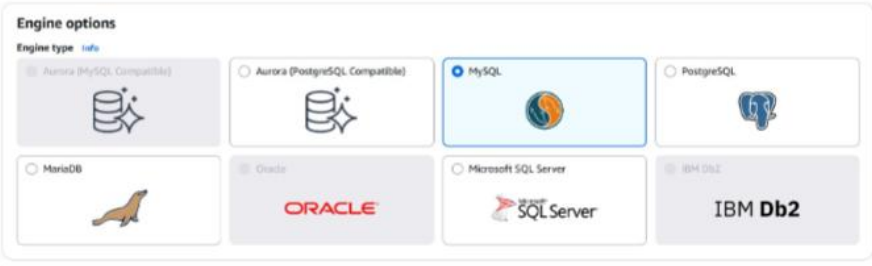
Click on create

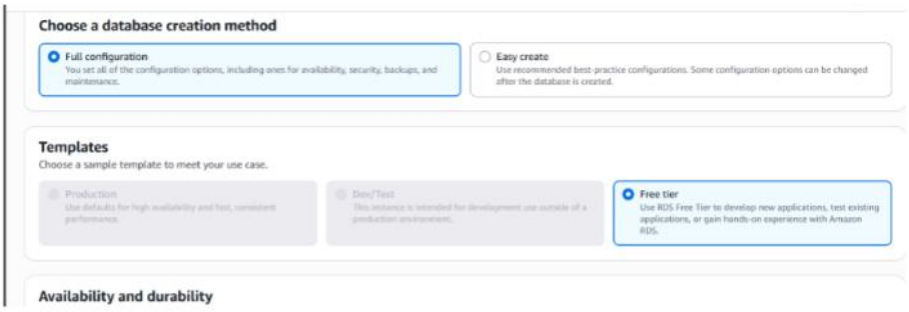
Create with full configuration

Create a database with custom configurations and enable features you want across a variety of engines including Aurora (compatible with MySQL & PostgreSQL), MySQL, PostgreSQL, MariaDB, Oracle, Microsoft SQL Server, and IBM Db2. Or use an existing database using restore from S3.

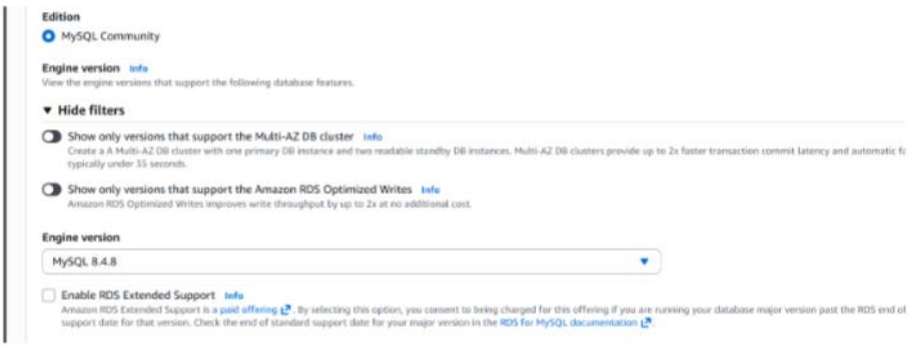
Create Restore from S3

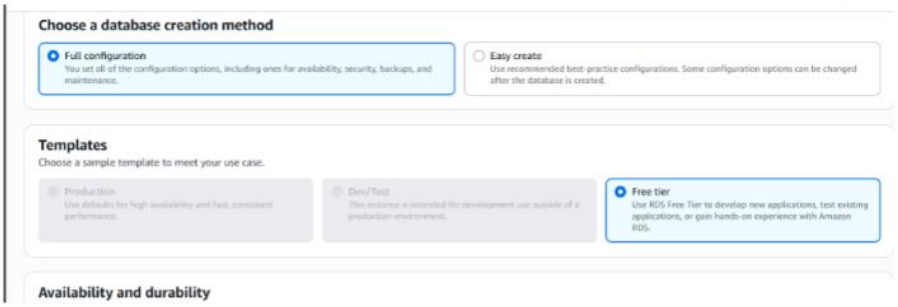
Click on create



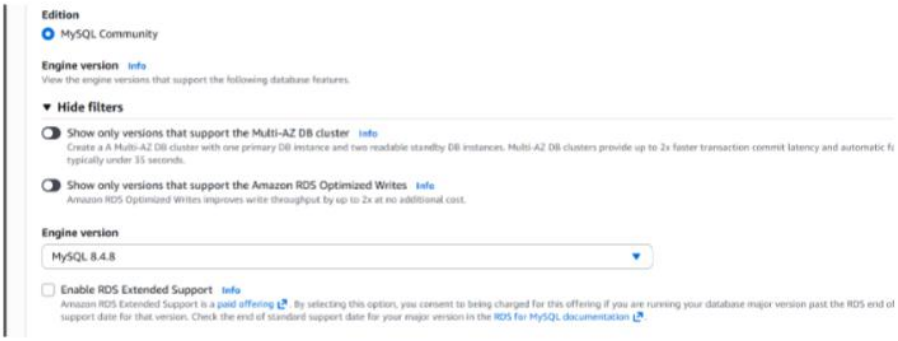


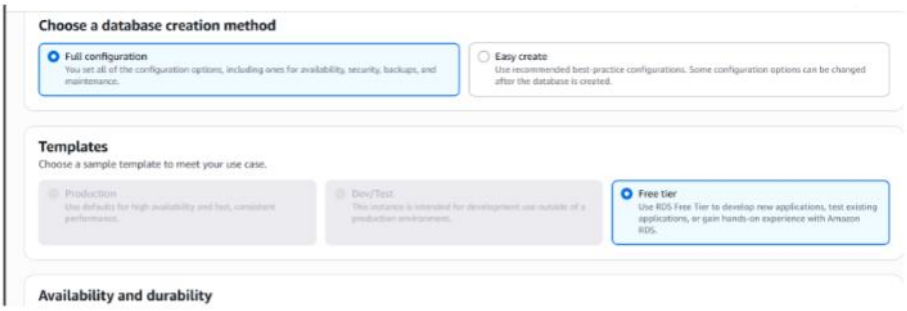
Select full configuration



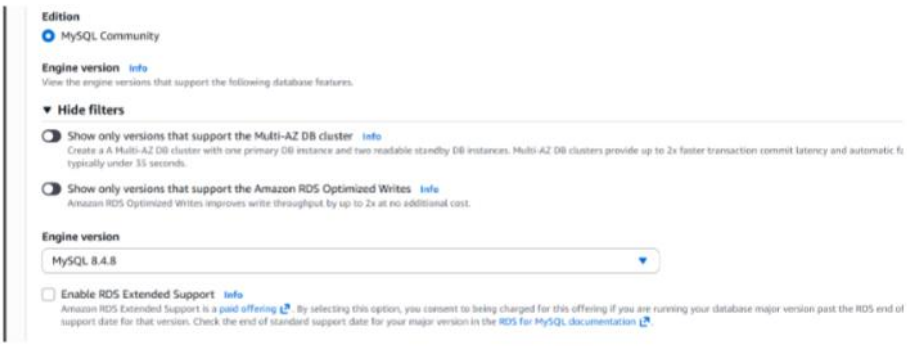


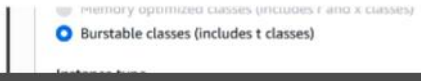
Select full configuration

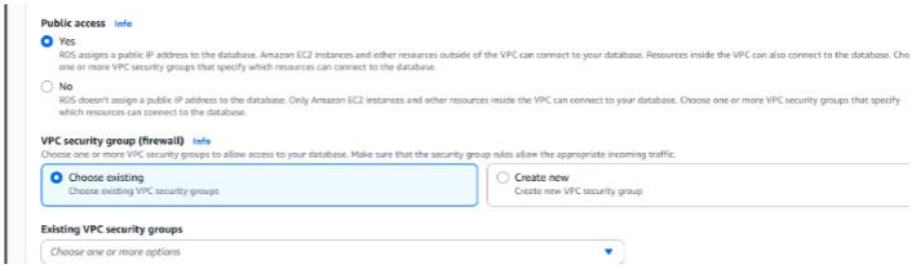




Select full configuration







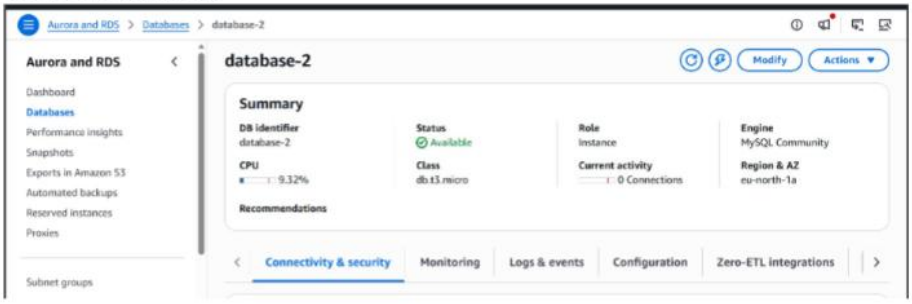
```
mysql> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
4 rows in set (0.00 sec)
```

8. Configure Multi AZ

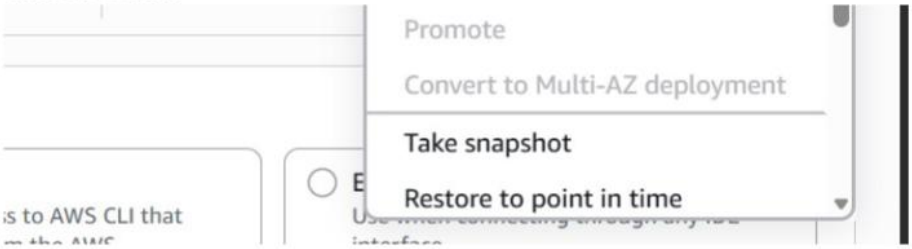
   Unable to configure multi AZ due to free tier restrictions

9. Take Backup of DB and restore the DB

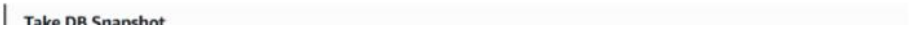
Go to the database

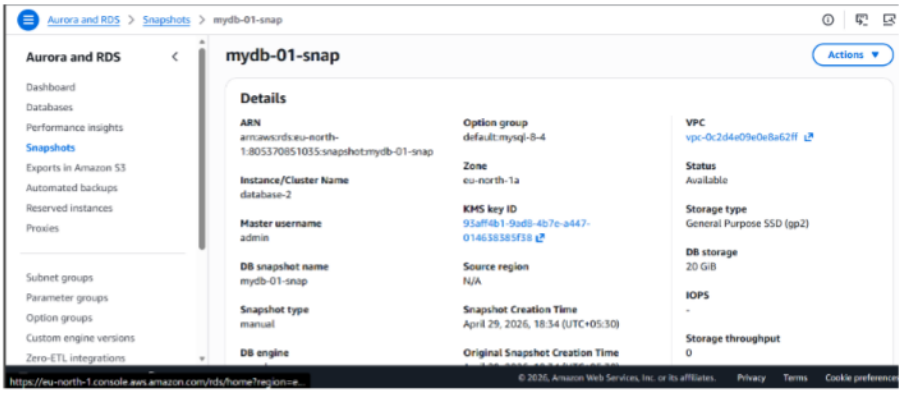


Click on actions

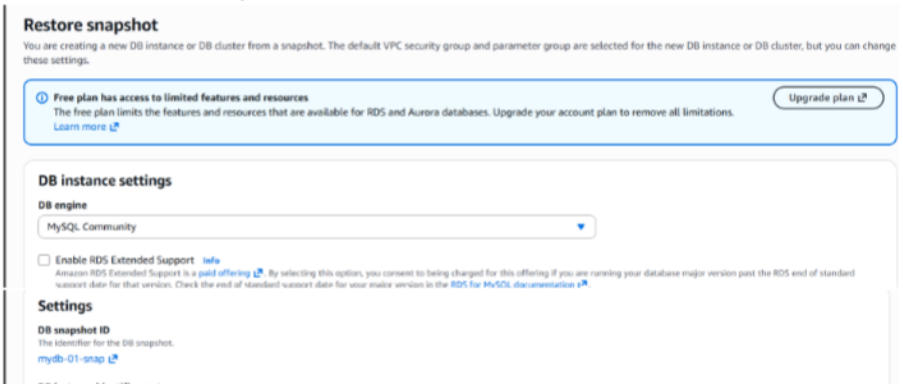


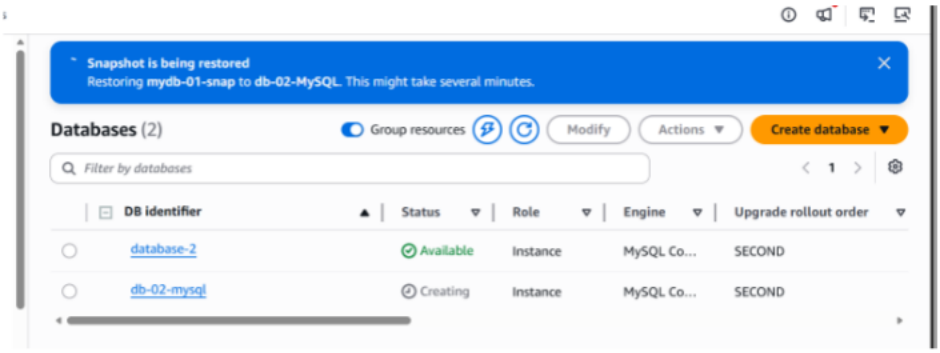
Click on take snapshot

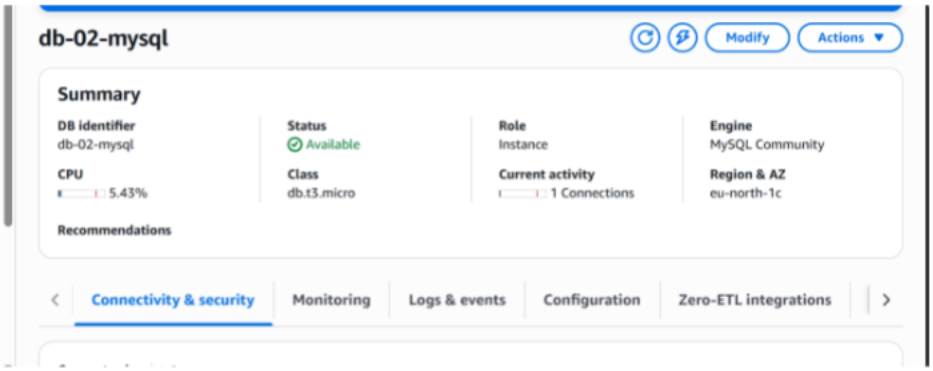




Snapshot is now created
To restore the snapshot click on actions
And then restore snapshot







The database is restored with the snapshot

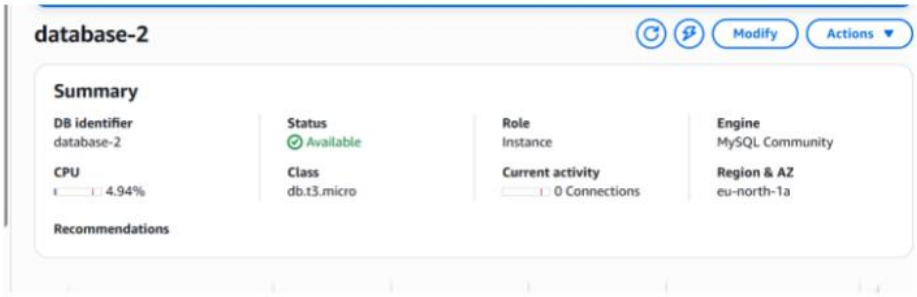
```
mysql> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
4 rows in set (0.02 sec)

mysql>
```

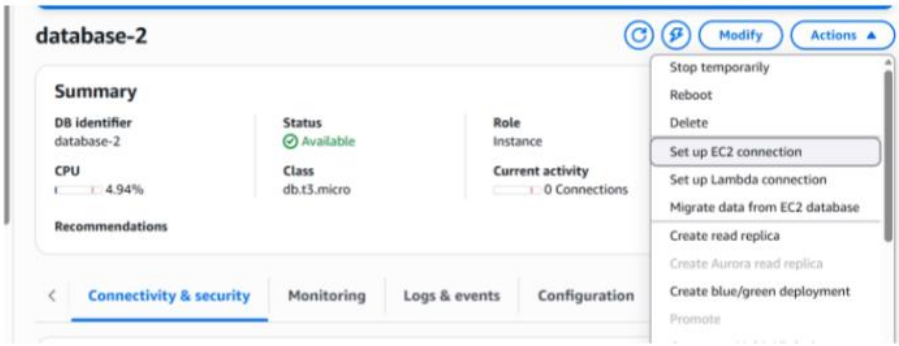
These are the databases available.

10. Create a Read Replica

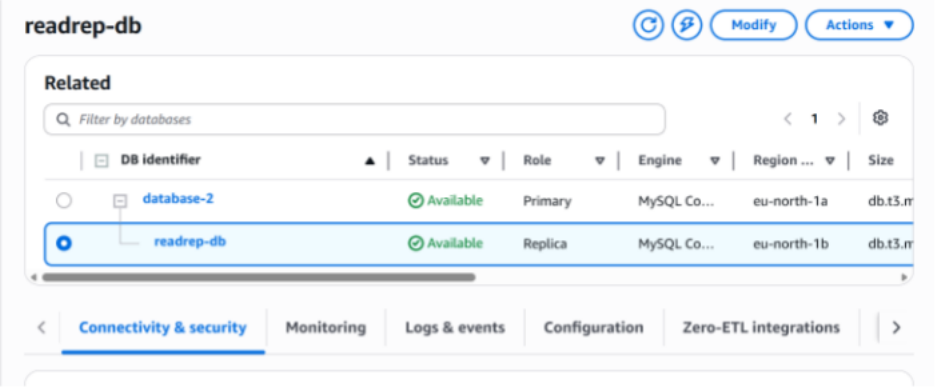
Go to the data base



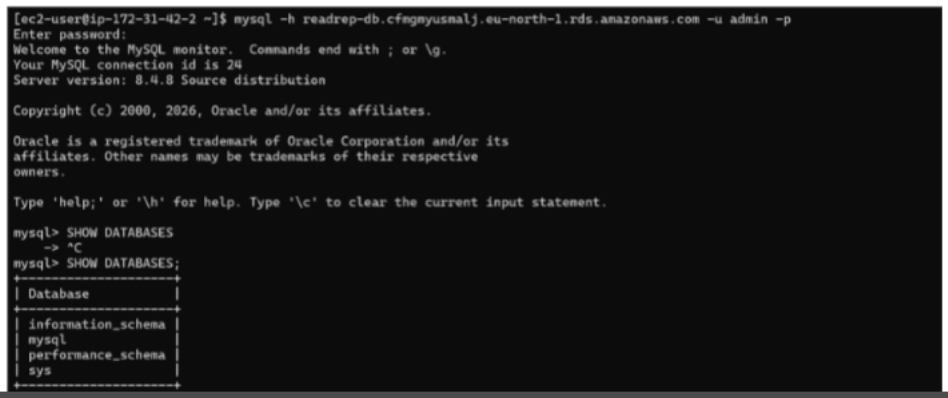
Click on actions



Click on create replica



Created the read replica



```
[ec2-user@ip-172-31-42-2 ~]$ mysql -h readrep-db.cfmgmyusmalj.eu-north-1.rds.amazonaws.com -u admin -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 24
Server version: 8.4.8 Source distribution

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> SHOW DATABASES
    -> ^C
mysql> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
```